STABILIZATION OF KERNELS OF POWERS OF A SELF-MAP MAKES THE KERNEL AND COKERNEL INVERSE SYSTEMS PRO-ISOMORPHIC IN THE DERIVED CATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains a commutative diagram typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so the diagram compiles exactly as in the source.

Editorial note. The proof cites no external \ref inside the counted span, so the assembly needs no external-reference stubs.

Editorial note. The author's closing phrase 'Some details omitted' concerns only the routine verification that the top-row complex is quasi-isomorphic to Coker(f^(n+c)); the argument lines are fully present.

Allowed standard prerequisites (editorial).

- Inverse systems and pro-isomorphism in the derived category. Stacks Project, Derived Categories of Modules (comparison of inverse systems as objects of D(A)). The conclusion of the lemma is a pro-isomorphism of two inverse systems in D(A); this is the ambient framework in which the two comparison arrows are formed.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $f : M \to M$ be a morphism of $\mathcal{A}$. If $M[f^n] = \mathrm{Ker}(f^n : M \to M)$ stabilizes, then the inverse systems*

$$(M \xrightarrow{f^n} M) \quad \text{and} \quad \mathrm{Coker}(f^n : M \to M)$$

are pro-isomorphic in $D(\mathcal{A})$.

Proof. There is clearly a map from the first inverse system to the second. Suppose that $M[f^c] = M[f^{c+1}] = M[f^{c+2}] = \ldots$. Then we can define an arrow of inverse systems in $D(\mathcal{A})$ in the other direction by the diagrams

$$\begin{array}{ccc} M/M[f^c] & \xrightarrow[f^{n+c}]{} & M \\ {\scriptstyle f^c}\downarrow & & \downarrow{\scriptstyle 1} \\ M & \xrightarrow{f^n} & M \end{array}$$

Since the top horizontal arrow is injective the complex in the top row is quasi-isomorphic to $\mathrm{Coker}(f^{n+c} : M \to M)$. Some details omitted. □

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